

TIC 373792319

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 16.198087 d
Epoch (t0) 2885.0829 BTJD
Duration 0.1700 d
Depth 8.978 % (89,784 ppm)
SNR 272.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

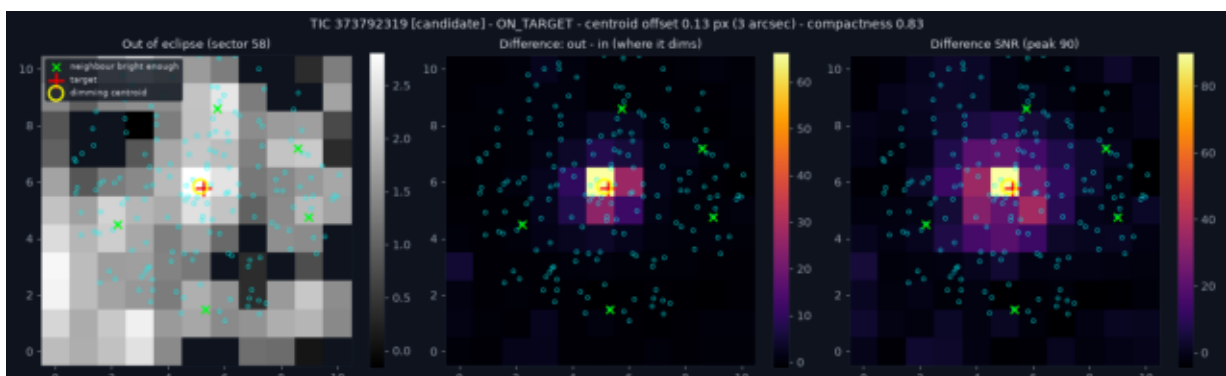
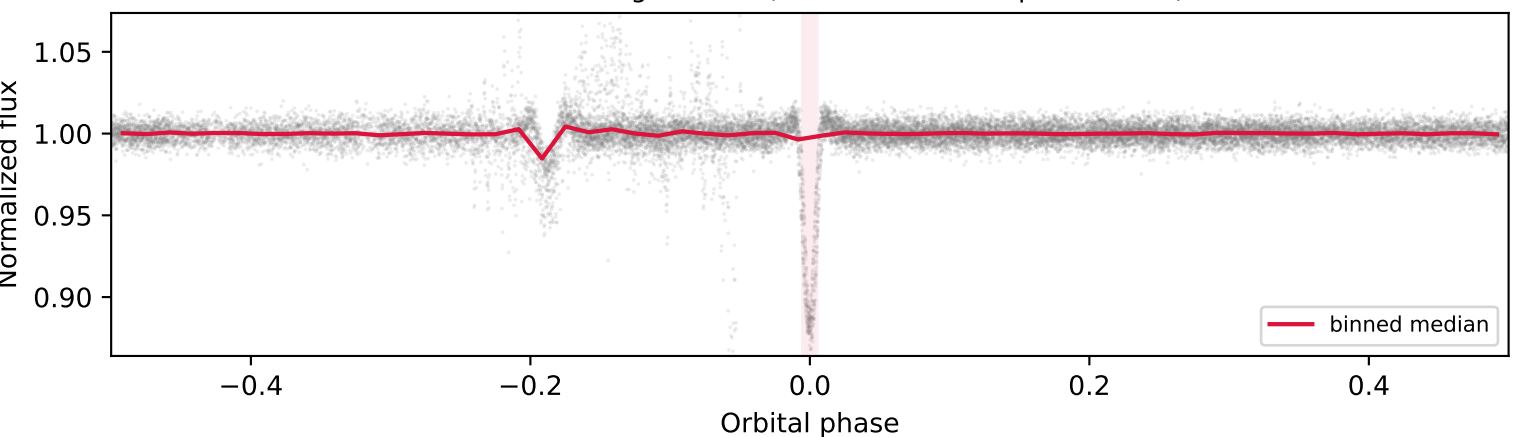
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 222.1 (gate ≥ 7.0)
Odd/even z-score 1.05 (consistent)
Secondary eclipse detected, phase -0.18, 7.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 58 light-curve dip in this sector: 157.3 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 90.22 (gate ≥ 5.0) Compactness 0.83 (gate ≥ 0.50)
Dimming centroid offset 0.13 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 373792377
2.84 px away, dTmag +0.49 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.